Hospital Performance & Patient Analytics

1. Project Overview

This project analyzes healthcare data to evaluate hospital performance, patient demographics, medical conditions, healthcare costs, and treatment patterns. Using a dataset containing approximately **56,000 patient records**, the analysis identifies trends in admissions, billing, insurance coverage, and hospital utilization. Python was used for data cleaning and exploratory analysis, PostgreSQL was used for SQL-based business analysis, and Power BI was used to build an interactive dashboard for visualization and decision-making.

2. Dataset Summary

The dataset contains approximately 56,000 patient records with information related to patient demographics, hospital admissions, diagnoses, billing, medications, insurance providers, and test results.

The main attributes can be grouped into the following categories:

- Patient Information: Name, Age, Gender, Blood Type, Age Group
- Medical Information: Medical Condition, Medication, Test Results
- Hospital Information: Hospital Name, Admission Type, Length of Stay
- Financial Information: Billing Amount, Insurance Provider
- Time Information: Date of Admission, Discharge Date, Season

During preprocessing, no major inconsistencies were found. Dates were converted into proper datetime format, column names were standardized, and additional features such as Age Group, Length of Stay, and Season were created to support further analysis.

3. Exploratory Data Analysis Using Python

Python was used to clean, prepare, and explore the dataset before loading it into PostgreSQL.

Data Loading

The dataset was imported into a pandas DataFrame for preprocessing and analysis.

Initial Exploration

Basic inspection techniques such as checking data types, summary statistics, duplicate values, and missing values were performed to understand the dataset structure.

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results	Length of Stay
0	Bobby Jackson	19	Female	AB+	Infections	2024-01-31	Matthew Smith	Northwestern Memorial Hospital	Blue Cross	2212.272701	328	Emergency	2024-02-07	Azithromycin	Normal	7
1	Leslie Terry	15	Female	B-	Flu	2019-08-20	Samantha Davies	UI Health (University of Illinois Hospital)	UnitedHealthcare	3185.161388	265	Emergency	2019-08-22	Tamiflu	Abnormal	2
2	Danny Smith	50	Female	A+	Cancer	2022-09-22	Tiffany Mitchell	UI Health (University of Illinois Hospital)	Blue Cross	72055.214065	205	Elective	2022-10-30	Cisplatin	Inconclusive	38
								UI Health								

Data Cleaning

- Removed duplicate records
- Converted admission and discharge dates into datetime format
- Standardized column names using snake_case formatting
- Verified data consistency across categorical columns

Feature Engineering

Several new variables were created to improve analysis:

- Age Group (Minor, Adult, Middle-aged, Senior)
- Length of Stay (calculated using admission and discharge dates)
- Season extracted from admission month

These additional features allowed more meaningful segmentation and trend analysis.

Database Integration

After preprocessing, the cleaned dataset was exported into a PostgreSQL database where SQL queries were executed for business analysis.

4. Data Analysis Using SQL

A series of SQL queries were executed in PostgreSQL to answer key healthcare and hospital management questions. These analyses provide insights into patient demographics, disease prevalence, healthcare costs, hospital performance, and treatment patterns.

Patient Distribution by Age and Gender

Patients were grouped according to Age Group and Gender to understand demographic distribution. This analysis helps identify the largest patient segments and supports healthcare planning for different age categories while comparing the proportion of male and female patients.

	age_group text	gender text	total_patients bigint
1	Adult	Female	6910
2	Adult	Male	6947
3	Middle-aged	Female	10960
4	Middle-aged	Male	10769
5	Minor	Female	2901
6	Minor	Male	2851
7	Senior	Female	7184
8	Senior	Male	6978

Most Common Medical Conditions

The number of patients diagnosed with each medical condition was calculated to identify the most prevalent diseases. Understanding disease frequency enables hospitals to prioritize healthcare resources, staffing, and preventive care programs.

	medical_condition text	patient_count bigint
1	Flu	7046
2	Diabetes	7005
3	Obesity	6994
4	Cancer	6940
5	Asthma	6908
6	Heart Disease	6900
7	Alzheimer's	6861
8	Infections	6846

Healthcare Costs by Medical Condition

Average billing amounts and total revenue generated by each medical condition were analyzed to determine which diseases contribute most to overall healthcare expenditure. This provides valuable insight into treatment costs and financial resource allocation.

	medical_condition text	average_bill numeric	total_bill numeric
1	Cancer	64537.09	447887395.35
2	Heart Disease	44913.43	309902646.64
3	Alzheimer's	32543.54	223281243.66
4	Diabetes	12503.19	87584835.23
5	Obesity	10055.56	70328608.00
6	Asthma	5025.35	34715105.99
7	Flu	2744.15	19335295.35
8	Infections	2747.57	18809847.06

Average Length of Hospital Stay

The average duration of hospitalization was calculated for each medical condition by measuring the difference between admission and discharge dates. Comparing the average length of stay helps identify conditions requiring prolonged treatment and greater hospital resource utilization.

	medical_condition text	avg_stay_days numeric
1	Alzheimer's	54.42
2	Cancer	36.54
3	Heart Disease	26.86
4	Diabetes	8.06
5	Obesity	5.97
6	Infections	5.52
7	Asthma	3.50
8	Flu	2.50

Insurance Provider Analysis

The number of patients covered by each insurance provider, along with the corresponding total billing amount, was analyzed. This helps evaluate the contribution of different insurance companies to hospital revenue and patient coverage.

	insurance_provider text	total_patients bigint	total_billing numeric
1	Cigna	11183	244135658.88
2	UnitedHealthcare	11142	242260181.26
3	Blue Cross	11134	243191828.30
4	Medicare	11038	241345747.32
5	Aetna	11003	240911561.51

Admission Type Distribution

Admissions were categorized into Emergency, Urgent, and Elective cases. The percentage share of each admission type was calculated to better understand patient admission patterns and assist hospitals in emergency preparedness and capacity planning.

	admission_type text	total_patients bigint	percentage numeric
1	Emergency	20104	36.22
2	Elective	13280	23.93
3	Urgent	11816	21.29
4	Routine	10300	18.56

Test Result Distribution

Laboratory test results were grouped into Normal, Abnormal, and Inconclusive categories. The percentage distribution of these outcomes provides an overview of patient health status and diagnostic trends across the dataset.

	test_results text	patients bigint	percentage numeric
1	Abnormal	27784	50.06
2	Normal	17453	31.45
3	Inconclusive	10263	18.49

Most Frequently Prescribed Medications

The frequency of each prescribed medication was calculated to identify the most commonly used drugs. This analysis assists hospitals in inventory management, pharmaceutical procurement, and understanding treatment preferences.

	medication text	prescription_count bigint
1	Metformin	4653
2	Zanamivir	2391
3	Methotrexate	2375
4	Glipizide	2362
5	Orlistat	2351
6	Insulin	2340
7	Aspirin	2337
8	Prednisone	2333
9	Oseltamivir	2329
10	Tamiflu	2326

Hospital Performance Analysis

Patient count and total billing amount were compared across hospitals to evaluate operational performance and revenue generation. This comparison helps assess hospital workload and financial contribution.

	hospital text	total_patients bigint	total_revenue numeric
1	Northwestern Memorial Hospital	14070	308689213.32
2	UI Health (University of Illinois Hospital)	13950	303837582.38
3	UChicago Medicine	13870	301281681.10
4	Loyola University Medical Center	13610	298036500.47

Monthly Admission Trends

Patient admissions were aggregated by month to examine how hospital admissions change throughout the year. Identifying monthly fluctuations helps hospitals forecast demand and optimize staffing and resource allocation.

	month_name text	total_admissions bigint
1	January	4692
2	February	4255
3	March	4672
4	April	4518
5	May	4599
6	June	4699
7	July	4812
8	August	4832
9	September	4546
10	October	4678
11	November	4548
12	December	4649

Seasonal Admission Patterns

Admissions were further analyzed by season to determine whether patient volume varies across different times of the year. Seasonal trend analysis supports long-term planning, resource management, and preparedness for periods of higher healthcare demand.

	season text	total_admissions bigint
1	Summer	14343
2	Spring	13789
3	Autumn	13772
4	Winter	13596

5. Dashboard in Power BI

An interactive two-page Power BI dashboard was developed to visualize healthcare performance and patient analytics.

Dashboard 1 – Trends

The first dashboard focuses on patient trends and healthcare performance.

Key performance indicators include:

- Total Patients
- Total Revenue

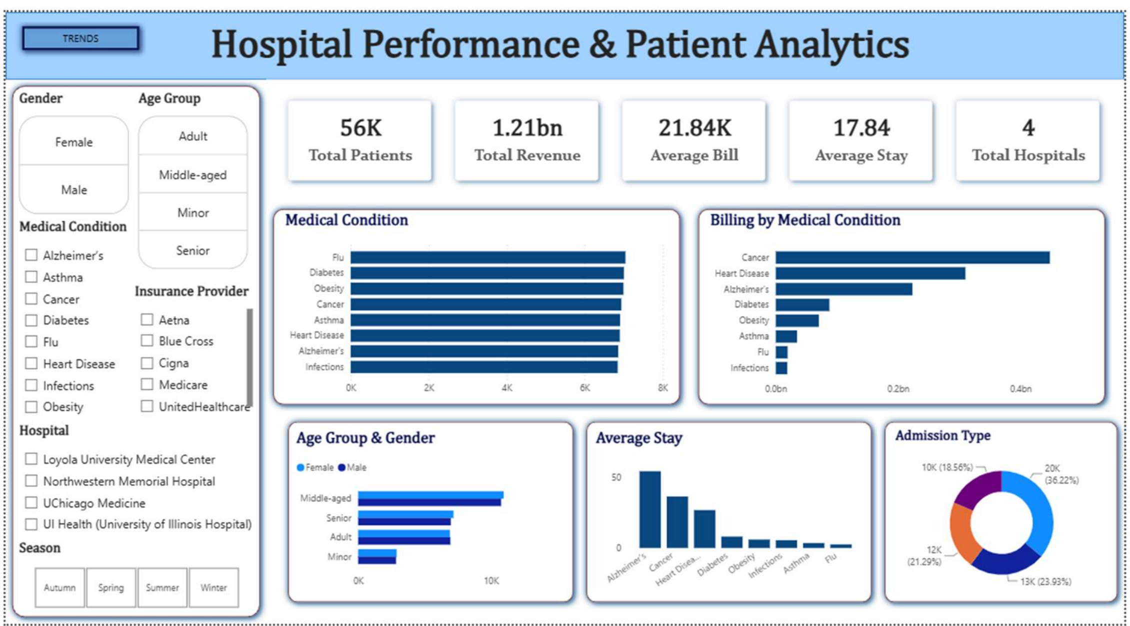
- Average Billing Amount
- Average Length of Stay
- Total Hospitals

Interactive visualizations include:

- Patient count by Medical Condition
- Billing Amount by Medical Condition
- Patient Distribution by Age Group and Gender
- Average Length of Stay by Medical Condition
- Admission Type Distribution

Users can filter the dashboard using:

- Gender
- Age Group
- Medical Condition
- Insurance Provider
- Hospital
- Season



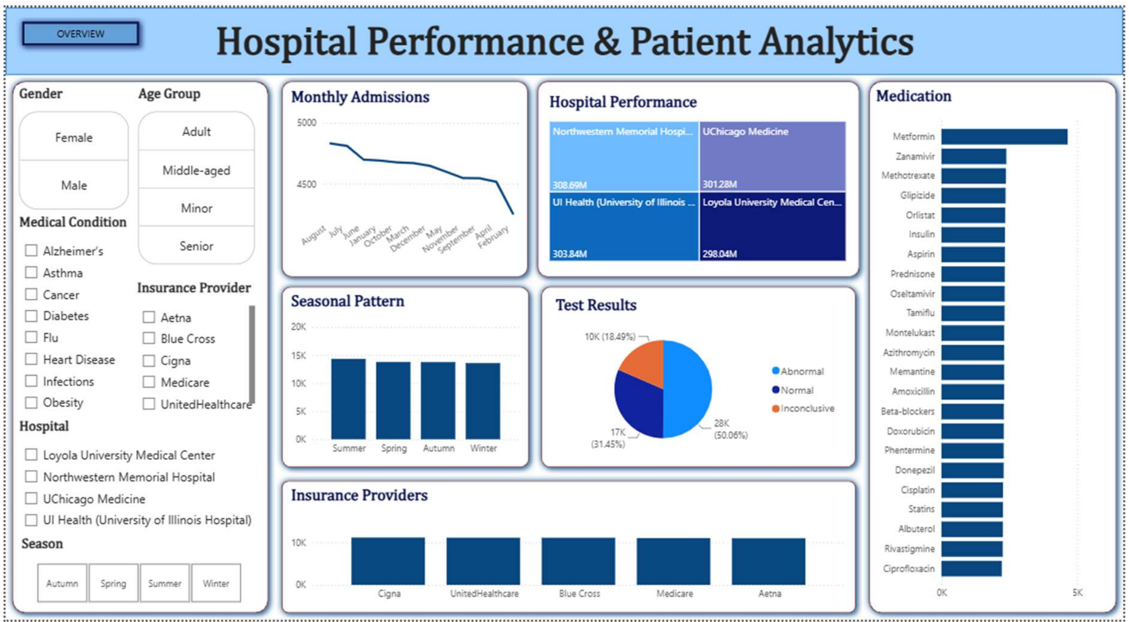
Dashboard 2 – Overview

The second dashboard provides operational insights into hospital performance.

Visualizations include:

- Monthly Admission Trend
- Hospital Revenue Comparison
- Seasonal Admission Pattern
- Test Result Distribution
- Insurance Provider Distribution
- Most Prescribed Medications

The dashboard enables users to explore patient demographics, healthcare costs, hospital efficiency, and treatment patterns through interactive filters and slicers.



6. Business Recommendations

Improve Preventive Healthcare

Medical conditions such as Diabetes and Heart Disease account for a significant proportion of patients. Preventive health programs and regular screening campaigns can help reduce long-term hospitalization.

Optimize Resource Allocation

Since Alzheimer's patients have the longest average hospital stays, hospitals should allocate additional beds, nursing staff, and specialized care resources to these departments.

Reduce Treatment Costs

Cancer contributes the highest billing amount. Cost optimization strategies, insurance collaborations, and efficient treatment planning can improve affordability without compromising care quality.

Strengthen Seasonal Planning

Patient admissions increase during certain seasons. Hospitals should adjust staffing levels and inventory planning according to expected seasonal demand.

Enhance Insurance Partnerships

Maintaining strong partnerships with major insurance providers can improve patient accessibility while ensuring consistent revenue streams.

Medication Inventory Management

Frequently prescribed medications should be prioritized in inventory planning to minimize shortages and improve treatment efficiency.

Monitor Hospital Performance

Since hospital revenues are relatively similar across institutions, administrators should compare operational efficiency, patient outcomes, and service quality to identify opportunities for improvement.